Blinkit Delivery Turnaround: Objective & North Star

A guide to diagnosing why orders are being delivered late

Objective :

Make Blinkit's 10-minute promise dependable as it scales, by finding where delivery time is spent across order wait, pick and pack, rider pickup and the last mile, and which of those parts Blinkit can control.

North Star Metric : On – Time Delivery Rate

$$\text{On-Time Delivery Rate} = \frac{\text{Orders delivered by promised time}}{\text{Eligible orders}} \times 100$$

Clock starts

order_placed_time

The moment the customer accepts the promise (payment confirmed). Starting here exposes queueing and dispatch delay.

Clock stops

actual_delivery_time

Rider marks the order delivered at the customer's location (handover, geo-confirmed where available).

On-time test

delivered_ts − placed_ts ≤ promised minutes

Promise is frozen at checkout with zero grace. Orders over 5 minutes late are tracked as 'severe late'.

Counts as NOT on time

Blinkit-caused cancellations (stock-out, no rider, store closed, system error) and any cancel made after the promised time has already passed.

Excluded from eligible orders

Customer cancels before picking starts, customer-side drop failures (unreachable, wrong address) and scheduled orders.

Guardrails : average and P90 delivery time, severe-late rate (over 5 min late), average promised time, credits per 100 orders, 30-day repeat rate and orders per rider-hour. Every metric is cut by city, store, store age, hour, day type and rain/event day.

Issue tree and metrics

ROOT	L1 STAGE	L2 METRIC, EQUATION & QUESTION	L3 METRIC & EQUATION	L3 ANALYTICAL QUESTION
On-Time Delivery Rate Delivery time **T = T1 + T2 + T3 + T4** Four stages, one after the other, none overlapping	**T1 Order wait** placed → picking starts T1 = pick_start_ts − placed_ts **Control: High**	**Picker load** [Store, Demand] Queue wait = pick_start_ts − placed_ts *Q: Do orders wait for a free picker before work starts?*	**Load per picker** Orders per picker-hour = Orders per hour ÷ Pickers on shift	*Is staffing keeping pace with order growth, especially 7-10 PM?*
			Queue length Avg queue = Σ orders waiting at placement ÷ Orders	*Which stores run a standing queue, and at which hours?*
		Demand spikes [Demand] Spike share = Orders in spike hours ÷ Total orders *Q: Is waiting caused by surges, not steady load?*	**Surge index** Surge = Orders per hour ÷ 4-week same-hour average	*Do rain, event and peak-hour surges explain the longest waits?*
	T2 Pick & pack picking starts → packed T2 = packed_ts − pick_start_ts **Control: High**	**Stock availability** [Store] Fill rate = Items available ÷ Items ordered *Q: Do stock-outs slow or break picking?*	**Stock-out cancels** Rate = Orders cancelled for stock-out ÷ Orders	*How often does missing stock end an order outright?*
		Picking efficiency [Store] Picking time = Σ (packed_ts − pick_start_ts) ÷ Orders *Q: Once started, how long does a store take, and why does it vary?*	**Minutes per item** Min per item = Picking time ÷ Item count	*Is slowness about basket size rather than the store?*
			Store-age gap Gap = Avg T2 (store < 90 days) − Avg T2 (store ≥ 90 days)	*Are newer stores slower after controlling for load?*
	T3 Rider pickup lag packed → picked up T3 = pickup_ts − packed_ts **Control: Medium**	**Rider supply** [Rider, Demand] Supply ratio = Riders logged in ÷ Orders per hour *Q: Is there enough rider capacity per order, hour by hour?*	**Peak supply dip** Dip = Supply ratio (7-10 PM) ÷ Supply ratio (2-5 PM)	*Does supply per order really fall in the evening peak?*
			Acceptance & idle Acceptance = Accepted ÷ Offered; Idle = Idle time ÷ Logged-in time	*Are riders present but rejecting or idle (allocation, not headcount)?*
		Store handoff [Rider, Store] Rider wait = pickup_ts − rider_arrival_ts *Q: How much rider and order time is lost at the store door?*	**Store-caused wait** Share = Trips (wait > 2 min, order unpacked) ÷ Trips	*Is the wait due to unpacked orders or early dispatch?*
			Packed-to-pickup lag (needs packed_ts) Lag = pickup_ts − packed_ts	*How often is an order packed and sitting with no rider?*
	T4 Last mile picked up → delivered T4 = delivered_ts − pickup_ts **Control: Low-Med**	**Distance & speed** [Rider] Distance per trip = Total km ÷ Trips *Q: Are stores serving wider catchments?*	**Transit speed** Speed = Km per trip ÷ T4 (min)	*Is lateness longer distance or slower riding?*
		Weather & events [Demand, Rider] Rain lift = Avg T4 (rain) ÷ Avg T4 (dry) *Q: How much last-mile delay is outside our control?*	**Event-day lift** Lift = Avg T4 (event) ÷ Avg T4 (normal day)	*What is the uncontrollable floor on delivery time?*

Data source tables are shown in [brackets]. T3 and T2 use picking time and rider wait as proxies until stage timestamps (pick_start_ts, packed_ts, rider_arrival_ts) are available.

Diagnostic Hypotheses & Priority Action Plan

1 Rider Supply Drop in the 7-10 PM Peak

Statement : Riders logged in per order fall sharply at peak, so pickup lag (T3) and late orders spike.

Test : Supply ratio = Riders logged in ÷ Orders per hour, and T3 minutes, by hour (7-10 PM vs 2-5 PM, same stores).

Reject If : Supply ratio and T3 are within ~10% of the afternoon, or evening lateness persists where the ratio is healthy.

Impact High | Data High | Actionable High | Score 9/9

2 Chronically Late Stores Driven by Capacity

Statement : A small share of stores, saturated on staff and queue, produce most late orders; city averages hide them.

Test : Share of late orders from the worst 10% and 20% of stores; late rate vs Orders per picker-hour and queue length.

Reject If : Late orders are spread evenly (worst 20% of stores < ~35% of late orders), or late rate does not rise with load.

Impact High | Data High | Actionable High | Score 9/9

3 Riders Wait at Stores for Unpacked Orders

Statement : Store-caused waiting is a material share of delivery time and hurts rider output.

Test : Avg rider wait per trip; % trips with wait > 2 min; correlation of wait with picking time and queue length.

Reject If : Median wait < ~1 min or < ~10% of delivery time, or wait is uncorrelated with picking time and queue.

Impact High | Data Medium (no packed_ts) | Actionable High | Score 8/9

4 Flat Delivery Time Promise

Statement : A flat 10-minute promise ignores distance and time of day, so late rate rises with distance and at peak.

Test : Std dev of promised minutes; late rate by distance band × hour band at the same promise.

Reject If : Promised time already varies with distance and hour, or late rate is flat across distance bands (< ~3 pt).

Impact Medium | Data Medium | Actionable High | Score 7/9

5 Complaints Jump When Orders Are Badly Late

Statement : Tickets and credits are non-linear: 1-2 minutes late rarely triggers a complaint, over 5-10 minutes does.

Test : Tickets per 100 orders and credit rate by minutes-late bucket: on time, 1-2, 3-5, 6-10, >10 min.

Reject If : Complaint rate rises roughly linearly with minutes late, or is already high at 1-2 minutes.

Impact Medium | Data High | Actionable Medium | Score 7/9

6 Newer Stores Are Slower

Statement : Stores under ~90 days old have longer picking time and a higher late rate than established stores.

Test : Picking time and late rate by store-age cohort (0-30, 31-90, 90+ days), controlling for city, load and staffing.

Reject If : The gap vanishes after controls (< ~2 pt), or variation within a cohort exceeds variation between cohorts.

Impact Medium | Data Medium (no open date) | Actionable Medium | Score 6/9

Ranking: score = impact + testability + actionability (High 3, Med 2, Low 1); ties broken by direct effect on late rate. **Run first:** split the 11 to 14 min rise into T1-T4 by city, store and hour, then test the top three in parallel. Thresholds (5 min, 10%, 3 pt) are starting values to calibrate on the data.